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ELECTRIC VEHICLE TRANSMISSION

Abstract

An electric vehicle transmission (**30**) for a vehicle includes a housing (**32**) defining a housing interior (**34**) and an electric motor (**40**) disposed in the housing interior (**34**). The electric motor (**40**) comprises a rotor (**42**) and a stator (**44**). The electric vehicle transmission (**30**) also includes an input shaft (**46**) disposed in the housing interior (**34**) and extending along a shaft axis. The input shaft (**46**) is rotatably coupled to the rotor (**42**) of the electric motor (**40**). The electric vehicle transmission (**30**) further includes a gear reduction assembly (**50**) disposed in the housing interior (**34**). The gear reduction assembly (**50**) is rotatably coupled to the input shaft (**46**) for delivering rotational torque to wheels of the vehicle. The housing (**32**) defines a first sump (**36**) for retaining oil and a second sump (**38**) separate from the first sump (**36**) for retaining oil. The second sump (**38**) and the first sump (**36**) are configured such that oil lubricates the gear reduction assembly (**50**) when flowing from the second sump (**38**) into the first sump (**36**).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application claims priority to and all the advantages of U.S. Provisional Patent Application No. 63/342,738, filed May 17, 2022, the entire contents of which are hereby incorporated by reference.
BACKGROUND OF THE DISCLOSURE
1. Field of the Disclosure
[0002] The present invention generally relates to an electric vehicle transmission.
2. Description of the Related Art
[0003] Conventional electric vehicle transmissions of a vehicle include a housing defining a housing interior and a sump for retaining oil, an electric motor including a rotor and a stator disposed in the housing interior, an input shaft disposed in the housing interior and extending along a shaft axis with the shaft being rotatably coupled to the rotor of the electric motor, and a gear reduction assembly disposed in the housing interior. The gear reduction assembly is rotatably coupled to the input shaft for delivering rotational torque to wheels of the vehicle. Oil is typically distributed throughout the electric vehicle transmission from the sump through the use of at least one pump. However, typical electric vehicle transmission using at least one pump to distribute oil throughout the electric vehicle transmission require the at least one pump to be larger, which occupies space in the vehicle and increases the cost of the electric vehicle assembly.
[0004] Accordingly, there remains a need for an improved electric vehicle transmission.
SUMMARY OF THE INVENTION
[0005] An electric vehicle transmission for a vehicle includes a housing defining a housing interior and an electric motor disposed in the housing interior. The electric motor comprises a rotor and a stator. The electric vehicle transmission also includes an input shaft disposed in the housing interior and extending along a shaft axis. The input shaft is rotatably coupled to the rotor of the electric motor. The electric vehicle transmission further includes a gear reduction assembly disposed in the housing interior. The gear reduction assembly is rotatably coupled to the input shaft for delivering rotational torque to wheels of the vehicle. The housing defines a first sump for retaining oil and a second sump separate from the first sump for retaining oil. The second sump and the first sump are configured such that oil lubricates the gear reduction assembly when flowing from the second sump into the first sump.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0006] Other advantages of the present disclosure will be readily appreciated, as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.

[0007] FIG. **1** is a perspective view of an electric vehicle transmission including a housing defining a housing interior;
[0008] FIG. **2** is a perspective view of a gear reduction assembly;
[0009] FIG. **3** is a front view of a gear reduction assembly;
[0010] FIG. **4** is a front view of the electric vehicle transmission including a housing defining a first sump and a second sump;
[0011] FIG. **5** is a perspective view of the electric vehicle transmission;
[0012] FIG. **6** is a perspective view of the electric vehicle transmission;
[0013] FIG. **7** is front view of a planetary gear set and a carrier of the electric vehicle transmission;
[0014] FIG. **8** is front view of a planetary gear set and another embodiment of the carrier of the electric vehicle transmission;
[0015] FIG. **9** is front view of a planetary gear set and another embodiment of the carrier of the electric vehicle transmission;
[0016] FIG. **10** is a cross-sectional view of the electric vehicle transmission, with the housing interior having an electric motor side and a gear reduction side; and
[0017] FIG. **11** is an exploded perspective view of the gear reduction assembly.

DETAILED DESCRIPTION OF THE INVENTION

[0018] With reference to the Figures, wherein like numerals indicate like parts throughout the several views, an electric vehicle transmission **30** for a vehicle is generally shown in FIGS. **1-11**. The electric vehicle transmission **30** includes a housing **32** defining a housing interior **34**. The electric vehicle transmission also includes an electric motor **40** disposed in the housing interior **34**. The electric motor **40** includes a rotor **42** and a stator **44**. The electric vehicle transmission **30** further includes an input shaft **46** disposed in the housing interior **34** and extending along a shaft axis SA. The input shaft **46** is rotatably coupled to the rotor **42** of the electric motor **40**. The electric vehicle transmission **30** typically includes a first output shaft **47**, and a second output shaft **49**. The input shaft **46** and the first output shaft **47** may be concentric shafts. The electric vehicle transmission **30** additionally includes a gear reduction assembly **50** disposed in the housing interior **34**. The gear reduction assembly **50** is rotatably coupled to the input shaft **46** for delivering rotational torque to wheels of the vehicle.

[0019] The housing **32** defines a first sump **36** for retaining oil and a second sump **38** separate from the first sump **36** for retaining oil. The housing **32** housing may include a housing protrusion **56** extending into the second sump **38** for further defining the second sump **38**. The second sump **38** and the first sump **36** are configured such that oil lubricates the gear reduction assembly **50** when flowing from the second sump **38** into the first sump **36**. Having the second sump **38** and the first sump **36** configured such that oil lubricates the gear reduction assembly **50** when flowing from the second sump **38** into the first sump **36** allows the electric vehicle transmission **30** to have a smaller pump to lubricating the electric motor **40**, as described in further detail below. Additionally, having the second sump **38** and the first sump **36** configured such that oil lubricates the gear reduction assembly **50** when flowing from the second sump **38** into the first sump **36** allows the lubrication of the gear reduction assembly **50** to be dependent on rotational speed of components of the electric vehicle transmission **30**, as described in further detail below.

[0020] In one embodiment, the second sump **38** and the first sump **36** are configured to passively control oil from flowing from the second sump **38** and the first sump **36**. In other words, the electric vehicle transmission **30** may be free of a valve, such as a solenoid valve, for actively controlling oil from flowing between the second sump **38** and the first sump **36**. Typically, the oil flows from the second sump **38** to the first sump **36** by way of gravity. To control the amount of oil flowing from the second sump **38** to the first sump **36**, an oil passageway **58** may be defined between the second sump **38** and the first sump **36**, with the flow area of the oil passageway **58** being sized based on the desired amount of flow of oil from the second sump **38** to the first sump **36**. Typically, the amount of oil that flows through the oil passageway **58** is less than the amount of oil that is bailed from the first sump **36** to the second sump **38**. Alternatively, the electric vehicle transmission **30** may include a valve **60**, such as a solenoid valve, disposed between the second sump **38** and said first sump to actively control oil from flowing between the second sump **38** and the first sump **36**.

[0021] The gear reduction assembly **50** may be further defined as a planetary gear set **62**. The planetary gear set **62** may include a plurality of pinion gears **64** disposed about the shaft axis SA. The plurality of pinion gears **64** may be defined as three pinion gears **64** and may be radially separated by **120** degrees from one another about the shaft axis SA.

[0022] The electric vehicle transmission **30** may include a carrier **66**, such as a disc, coupled to the gear reduction assembly **50**. When present, the carrier **66** is configured to bail (i.e., deliver) oil from the first sump **36** to the second sump **38** during rotation of the gear reduction assembly **50**. In one embodiment, the carrier **66** has a circular configuration. The carrier **66** may have a first carrier surface **68** facing a first direction with respect to the shaft axis SA, and a second carrier surface **70** facing a second direction opposite the first direction with respect to the shaft axis SA. The first and second carrier surfaces **68**, **70** may have a flat configuration. Having a flat configuration allows the carrier **66** to rotate through the oil in the first sump with minimal drag losses, all while still allowing oil to dispose on the first and second carrier surfaces **68**, **70** and to be delivered to the second sump **38** due to rotation of the carrier **66**. The first and second carrier surfaces **68**, **70** may be parallel to one another. It is to be appreciated that the first and second carrier surfaces **68**, **70** may have elevations and/or indentations for aiding in the bailing of oil from the first sump **36** to the second sump **38** during rotation of the carrier **66**. Whether the first and second carrier surfaces **68**, **70** are flat, or have indentations and/or elevations, is dependent balancing the desire for reduced drag of the carrier **66** through the oil in the first sump **36** against the amount of oil that needs to be bailed from the first sump **36** to the second sump **38**.

[0023] The pinion gears **64** may define a pinion diameter PD. The carrier **66** may define a carrier diameter CD. Typically, the carrier diameter CD is greater than the pinion diameter PD. It is to be appreciated that the carrier **66** may have an increased diameter portion to further allow the carrier **66** to enter the oil disposed in the first sump **36** to then bail the oil into the second sump **38** during rotation of the carrier **66**. In addition to be configured to bail the oil from the first sump **36** to the second sump **38**, the carrier **66** also provides additional support to the pinion gears **64** such that deflection of components of the electric vehicle transmission **30**, in particular the gear reduction assembly **50**, is reduced. To this end, when the carrier **66** is present, the carrier **66** is able to move oil from the first sump **36** to the second sump **38** and reduce spin losses of the electric vehicle transmission **30**.

[0024] The housing interior **34** may be divided into an electric motor side **74** and a gear reduction side **76**, as best shown in FIG. **10**. Typically, the electric vehicle transmission **30** includes a pump for circulating oil in the electric motor side **74** of the housing interior **34**. The electric vehicle transmission **30** may be free of a pump for circulating oil in the gear reduction side **76** of the housing interior **34**. Due to the requirement to cool the electric motor **40**, the electric motor side **74** of the housing interior **34** typically requires a higher flow and circulation of oil to cool the electric motor **40**. The pump circulates the oil and cools the oil through a heat exchanger of the electric vehicle transmission **30**. When the carrier **66** is present, the gear reduction side **76** of the housing interior **34** typically does not require a pump due to the carrier **66** bailing oil from the first sump **36**, to the second sump **38**, and allowing gravity to draw oil from the second sump **38** toward the first sump **36** and lubricating the gear reduction assembly **50**. To help with directing the oil throughout the gear reduction assembly **50** and to other components of the electric vehicle transmission **30**, the electric vehicle transmission **30** may include an oil distribution assembly **78** disposed in the housing interior **34**. Typically, the oil distribution assembly **78** is disposed between the second sump **38** and the first sump **36** such that oil flows from the second sump **38**, to the oil distribution assembly **78**, and to the first sump **36** after the oil lubricates the gear reduction assembly **50** and other components of the electric vehicle transmission **30**, such as bearings, gears, and bushings.

[0025] To balance the amount of oil in the electric motor side **74** and the gear reduction side **76** of the housing interior, the housing **32** in the second sump **38** may define an upper overfill bleed orifice **82** for allowing oil to flow between the electric motor side **74** and the gear reduction side **76**. Similarly, the housing **32** in the first sump **36** may define a lower overfill bleed orifice **82** for allowing oil to flow between the electric motor side **74** and the gear reduction side **76**.

[0026] Embodiment 1: An electric vehicle transmission for a vehicle, comprising: [0027] a housing defining a housing interior; [0028] an electric motor disposed in said housing interior, wherein said electric motor comprises a rotor and a stator; [0029] an input shaft disposed in said housing interior and extending along a shaft axis, wherein said input shaft is rotatably coupled to said rotor of said electric motor; [0030] a gear reduction assembly disposed in said housing interior, wherein said gear reduction assembly is rotatably coupled to said input shaft for delivering rotational torque to wheels of the vehicle; [0031] wherein said housing defines a first sump for retaining oil and a second sump separate from said first sump for retaining oil; [0032] wherein said second sump and said first sump are configured such that oil lubricates said gear reduction assembly when flowing from said second sump into said first sump.

[0033] Embodiment 2: The electric vehicle transmission as set forth in embodiment 1, wherein said second sump and said first sump are configured to passively control oil from flowing between said second sump and said first sump.

[0034] Embodiment 3: The electric vehicle transmission as set forth in embodiment 1 further comprising a valve disposed between said second sump and said first sump to actively control oil from flowing between said second sump and said first sump.

[0035] Embodiment 4: The electric vehicle transmission as set forth in any one of the preceding embodiments, wherein said gear reduction assembly is further defined as a planetary gear set.

[0036] Embodiment 5: The electric vehicle transmission as set forth in embodiment 4, wherein said planetary gearset comprises a plurality of pinion gears disposed about said shaft axis.

[0037] Embodiment 6: The electric vehicle transmission as set forth in embodiment 5, wherein said plurality of pinion gears is further defined as three pinion gears.

[0038] Embodiment 7: The electric vehicle transmission as set forth in embodiment 6, wherein said three pinion gears are radially separated by 120 degrees from one another about said shaft axis.

[0039] Embodiment 8: The electric vehicle transmission as set forth in any one of the preceding embodiments further comprising a carrier coupled to said gear reduction assembly, wherein said carrier is configured to bail oil from said first sump to said second sump during rotation of said gear reduction assembly.

[0040] Embodiment 9: The electric vehicle transmission as set forth in embodiment 8, wherein said carrier has a circular configuration.

[0041] Embodiment 10: The electric vehicle transmission as set forth in any one of embodiments 8 and 9, wherein said carrier has a first carrier surface facing a first direction with respect to said shaft axis, and a second carrier surface facing a second direction opposite said first direction with respect to said shaft axis, wherein said first and second carrier surfaces have a flat configuration.

[0042] Embodiment 11: The electric vehicle transmission as set forth in any one of embodiments 8 and 9, wherein said carrier has a first carrier surface facing a first direction with respect to said shaft axis, and a second carrier surface facing a second direction opposite said first direction with respect to said shaft axis, wherein said first and second carrier surfaces have elevations and/or indentations.

[0043] Embodiment 12: The electric vehicle transmission as set forth in any one of embodiments 8 and 9, wherein said carrier has a first carrier surface facing a first direction with respect to said shaft axis, and a second carrier surface facing a second direction opposite said first direction with respect to said shaft axis, wherein at least one of said first and second carrier surfaces have elevations and/or indentations for aiding in the bailing of oil from said first sump to said second sump during rotation of said carrier.

[0044] Embodiment 13: The electric vehicle transmission as set forth in any one of the preceding embodiments, wherein said housing comprises a housing protrusion extending into said second sump for further defining said second sump.

[0045] Embodiment 14: The electric vehicle transmission as set forth in any one of the preceding embodiments, wherein said housing interior is divided into an electric motor side and a gear reduction side.

[0046] Embodiment 15: The electric vehicle transmission as set forth in embodiment 14, wherein said housing in said second sump defines an upper overfill bleed orifice for allowing oil to flow between said electric motor side and said gear reduction side.

[0047] Embodiment 16: The electric vehicle transmission as set forth in any one of embodiments 14 and 15, wherein said housing in said first sump defines a lower overfill bleed orifice for allowing oil to flow between said electric motor side and said gear reduction side.

[0048] Embodiment 17: The electric vehicle transmission as set forth in any one of embodiments 14-16 further comprising a pump for circulating oil in said electric motor side of said housing interior.

[0049] Embodiment 18: The electric vehicle transmission as set forth in any one of embodiments 14-17 being free of a pump for circulating oil in said gear reduction side of said housing interior.

[0050] Embodiment 19: The electric vehicle transmission as set forth in any one of the preceding embodiments further comprising an oil distribution assembly disposed in said housing interior.

[0051] Embodiment 20: The electric vehicle transmission as set forth in embodiment 19, wherein said oil distribution assembly is disposed between said second sump and said first sump such that oil flows from said second pump, to said oil distribution assembly, and to said first sump after the oil lubricates the gear reduction assembly.

Claims

1. An electric vehicle transmission for a vehicle, comprising: a housing defining a housing interior; an electric motor disposed in said housing interior, wherein said electric motor comprises a rotor and a stator; an input shaft disposed in said housing interior and extending along a shaft axis, wherein said input shaft is rotatably coupled to said rotor of said electric motor; a gear reduction assembly disposed in said housing interior, wherein said gear reduction assembly is rotatably coupled to said input shaft for delivering rotational torque to wheels of the vehicle; wherein said housing defines a first sump for retaining oil and a second sump separate from said first sump for retaining oil; wherein said second sump and said first sump are configured such that oil lubricates said gear reduction assembly when flowing from said second sump into said first sump; and wherein said housing interior is divided into an electric motor side and a gear reduction side.

2. The electric vehicle transmission as set forth in claim 1, wherein said second sump and said first sump are configured to passively control oil from flowing between said second sump and said first sump.

3. The electric vehicle transmission as set forth in claim 1 further comprising a valve disposed between said second sump and said first sump to actively control oil from flowing between said second sump and said first sump.

4. The electric vehicle transmission as set forth in claim 1, wherein said gear reduction assembly is further defined as a planetary gear set.

5. The electric vehicle transmission as set forth in claim 4, wherein said planetary gearset comprises a plurality of pinion gears disposed about said shaft axis.

6. The electric vehicle transmission as set forth in claim 5, wherein said plurality of pinion gears is further defined as three pinion gears.

7. The electric vehicle transmission as set forth in claim 6, wherein said three pinion gears are radially separated by 120 degrees from one another about said shaft axis.
8. The electric vehicle transmission as set forth in claim 1, further comprising a carrier coupled to said gear reduction assembly, wherein said carrier is configured to bail oil from said first sump to said second sump during rotation of said gear reduction assembly.
9. The electric vehicle transmission as set forth in claim 8, wherein said carrier has a circular configuration.
10. The electric vehicle transmission as set forth in claim 8, wherein said carrier has a first carrier surface facing a first direction with respect to said shaft axis, and a second carrier surface facing a second direction opposite said first direction with respect to said shaft axis, wherein said first and second carrier surfaces have a flat configuration.
11. The electric vehicle transmission as set forth in claim 8, wherein said carrier has a first carrier surface facing a first direction with respect to said shaft axis, and a second carrier surface facing a second direction opposite said first direction with respect to said shaft axis, wherein said first and second carrier surfaces have elevations and/or indentations.
12. The electric vehicle transmission as set forth in claim 8, wherein said carrier has a first carrier surface facing a first direction with respect to said shaft axis, and a second carrier surface facing a second direction opposite said first direction with respect to said shaft axis, wherein at least one of said first and second carrier surfaces have elevations and/or indentations for aiding in the bailing of oil from said first sump to said second sump during rotation of said carrier.
13. The electric vehicle transmission as set forth in claim 1, wherein said housing comprises a housing protrusion extending into said second sump for further defining said second sump.
14. The electric vehicle transmission as set forth in claim 1, wherein said housing in said second sump defines an upper overfill bleed orifice for allowing oil to flow between said electric motor side and said gear reduction side.
15. The electric vehicle transmission as set forth in claim 1, wherein said housing in said first sump defines a lower overfill bleed orifice for allowing oil to flow between said electric motor side and said gear reduction side.
16. The electric vehicle transmission as set forth in claim 1, further comprising a pump for circulating oil in said electric motor side of said housing interior.
17. The electric vehicle transmission as set forth in claim 1, being free of a pump for circulating oil in said gear reduction side of said housing interior.
18. The electric vehicle transmission as set forth in claim 1, further comprising an oil distribution assembly disposed in said housing interior.
19. The electric vehicle transmission as set forth in claim 18, wherein said oil distribution assembly is disposed between said second sump and said first sump such that oil flows from said second sump, to said oil distribution assembly, and to said first sump after the oil lubricates the gear reduction assembly.